Abhishek **Datta**

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Summary

Data Engineer with 5+ years of experience designing large-scale data platforms and distributed ETL systems using Python, PySpark, Apache Iceberg, Databricks, and Snowflake. Passionate about building production-grade software systems spanning computer vision, security, cloud platforms, and backend engineering

Experience

Senior Data Engineer May 2026 - Present
IBM India

- Developed large-scale data engineering solutions on an on-premise Cloudera Data Platform (CDP Bare Metal) using Python, PySpark, Pandas, Hive SQL, Impala SQL, Apache Iceberg, Bash and Linux for one of India's largest banking institutions
- Engineered and automated a metadata-driven migration framework for 75,000+ tables and 12,000+ materialized and non-materialized views from IBM DB2 to Apache Iceberg on Cloudera using Python-based syntax transformation and dictionary-driven automation, ensuring compatibility with modern Iceberg architecture while preserving existing business logic and requirements

Data Engineer June 2022 - May 2026
LTM (LTIMindtree) India

- Designed and implemented end-to-end data systems on Databricks and Snowflake, including architecture, pipeline development, orchestration, and monitoring
- Developed scalable ETL pipelines using Python, PySpark, SQL, Databricks, and Snowflake to handle near real-time data ingestion and transformation
- Automated CI/CD workflows along with DevOps team using Azure DevOps and Harness, reducing deployment time by 40%
- Optimized data pipelines by improving Spark transformations, SQL queries, and resource utilization, reducing execution time, improving performance and lowering costs on Databricks and consistently delivered projects ahead of schedule, receiving recognition for solution quality, delivery speed, and technical ownership

PAT August 2020 - May 2021
Cognizant India

- Enhanced Salesforce Cloud workflows and supported enterprise data integration using Salesforce, Informatica, and MySQL

Personal Projects

Project ZeroTrust (June 2026 - Present)
Python • FastAPI • OpenCV • NumPy • Angular • Git

- Built a computer vision-based real-time security application that continuously authenticates user presence and automatically locks the workstation upon unauthorized access detection, with configurable security policies
- Designed a watchdog-based monitoring architecture to detect and handle abnormal, unauthorized, or forced process termination and ensure continuous protection through automatic self-recovery
- Developed authentication-guarded administrative operations to prevent unauthorized shutdown, configuration changes, or disabling of security features, and engineered self-protection mechanisms that detect process tampering and trigger automatic workstation lockdown upon security policy violations
- Integrated the image embedding pipeline with in-memory embedding storage to deliver low-latency, real-time identity verification, optimized for low-specification hardware with minimal CPU and memory overhead

Masterclass Booking System (May 2025 - Present)
Java • Spring Boot • PostgreSQL • Razorpay • Docker • GCP • REST API • Angular • Git

- Built a full-stack booking and student management platform enabling class registration, scheduling/rescheduling, Razorpay payment processing, recurring billing, and instructor–student workflow automation using RESTful APIs
- Leveraged Google Cloud services, including IAM, Service Accounts, Secret Manager, Google Calendar API and implemented automated email workflows for OTP verification, calendar invitations, registration confirmation, payment receipts, and class scheduling notifications

Education

University of Engineering & Management, Kolkata India
Bachelor of Technology in Computer Science & Engineering May 2016 - May 2020

Technical Skills

Languages: Python, Advanced SQL, Java
Data Engineering: PySpark, Hive, Impala, Apache Iceberg, Snowflake, Databricks
Cloud: Cloudera CDP, Azure, GCP
Tools: Docker, Git, Linux

Certifications & Badges

Databricks Certified Data Engineer Associate • **Microsoft** Azure Fundamentals • **Hackerrank** SQL Advanced
IBM Agentic & Generative AI Developer • **IBM** Enterprise Design Thinking Practitioner